

ShaftCoupledTurbine

$$H = H_{\text{in}} - H_{\text{out}}$$

$$y = u_{\text{gate}}$$

$$Q = K_q y \sqrt{H}$$

$$P_h = \rho g \eta Q H$$

$$\tau_{\text{shaft}} \omega_{\text{shaft}} = -P_h$$

RotationalPort

LumpedShaft

state: ω

$$J \dot{\omega} = \tau_{\text{drive}} + \tau_{\text{load}} - d(\omega - \omega_0)$$

$$\omega_{\text{drive}} = \omega_{\text{load}} = \omega$$

RotationalPort

SMIBGenerator

state: δ

$$\dot{\delta} = \omega_m - \omega_{\text{grid}}$$

$$P_e = P_{\text{max}} \sin \delta$$

$$P_e = \eta \tau_{\text{shaft}} \omega_{\text{shaft}}$$

$$P_{\text{grid}} = -P_e$$

connector torque is positive *into* each component;
generation therefore appears with the appropriate sign at each connected port.